

# Split Conformal Prediction for Regression

A Tutorial

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# Outline

Motivation

Split Conformal Prediction

Proof of Marginal Coverage Guarantee

Finite Sample Bound

Test-conditional Coverage and Split Localized Conformal Prediction

Appendix

# Download



Python notebook for tutorial



Lecture notes

# Outline

## Motivation

## Split Conformal Prediction

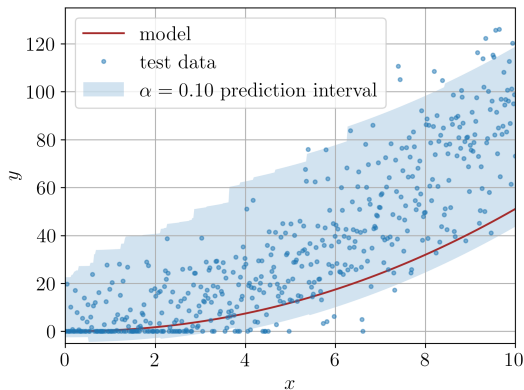
## Proof of Marginal Coverage Guarantee

## Finite Sample Bound

## Test-conditional Coverage and Split Localized Conformal Prediction

## Appendix

## What is the goal?



- Given a statistical model  $\hat{f} : x \mapsto y$ , we want to quantify the probability that the true value  $y$  for an input  $x$  is within a prediction set  $\mathcal{C}(x)$ .
- What would be your favourite method to address this task?
- Does this approach give you rigorous guarantees? That is, can you guarantee that the predictions will be in the prediction set with a probability of  $(1 - \alpha)$ ?

## Problem formulation

Given

- distribution  $P$  over  $\mathcal{X} \times \mathcal{Y}$ ,
- fitted model  $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$ ,
- miscoverage level  $\alpha \in \mathbb{R}$ ,
- calibration data  $\{X_i, Y_i\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} P$ ,

construct a function  $\mathcal{C} : \mathcal{X} \rightarrow 2^{\mathcal{Y}}$  that generates a prediction interval  $\mathcal{C}(X_{\text{test}})$  such that

$$\mathbb{P}(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}})) \geq 1 - \alpha$$

for any test sample  $(X_{\text{test}}, Y_{\text{test}}) \sim P$ .

## Coverage and conditional coverage

- There is an important difference between (marginal) coverage

$$\mathbb{P}(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}})) \geq 1 - \alpha$$

and conditional coverage:

$$\mathbb{P}(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}}) \mid X_{\text{test}}) \geq 1 - \alpha$$

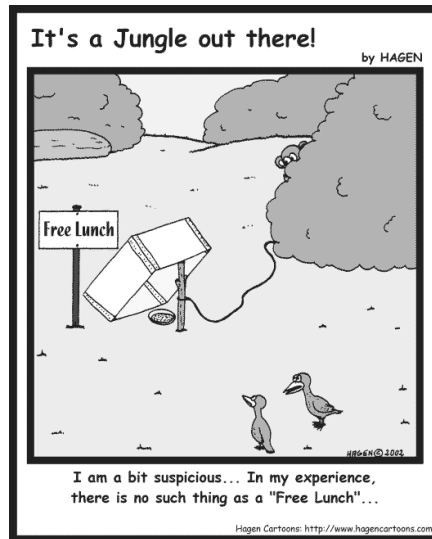
- Conditional coverage is difficult to achieve, in the continuous case in general impossible.
- Important: Do guarantees hold in expectation over all possible calibration data sets  $\mathcal{D}_{\text{cal}}$  or for a concrete finite  $\mathcal{D}_{\text{cal}}$ ?

## No free lunch

It is not cheap to get rigorous uncertainty estimates.

You have to pay either by

- strong accurate assumption on the hypothesis class, the data generating distribution and/or the algorithm, or
- having i.i.d. calibration data.





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## Empirical cumulative distribution and quantiles

- Random variables and their realizations are denoted by uppercase and lowercase letters, respectively.
- $F_X(x) = \mathbb{P}(X \leq x)$ : cumulative distribution function (CDF) of real-valued  $X$ .
- $\hat{F}_X(z) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{x_i \leq z\}$ : empirical CDF based on realizations  $\{x_i\}_{i=1}^n$  of  $X$ .
- $\mathcal{Q}_\tau(F_X) = \inf\{x : \tau \leq F_X(x)\}$ :  $\tau$ -quantile of a CDF  $F_X$ .
- Empirical  $\tau$ -quantile of a collection  $\{z_i\}_{i=1}^n$  using nearest-rank definition:

$$\widehat{\mathcal{Q}}_\tau(\{z_i\}_{i=1}^n) = \inf \left\{ z' : \tau \leq \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{z_i \leq z'\} \right\} ,$$

which is the  $\lceil \tau \cdot n \rceil$ -th value in the list of  $\{z_i\}_{i=1}^n$  values sorted in increasing order:

$$\widehat{\mathcal{Q}}_{0.5}(\{1, 2, 3, 4, 5\}) = 3 \quad , \quad \widehat{\mathcal{Q}}_{0.5}(\{1, 2, 3, 4\}) = 2 \quad , \quad \widehat{\mathcal{Q}}_{0.25}(\{1, 2, 3, \dots, 100\}) = 25$$

## Basic idea of split conformal prediction I

A standard (machine learning) way to estimate the accuracy of a predictive model:

- To estimate the expected accuracy of a model, we estimate its accuracy on i.i.d. validation data not used during training.
- In expectation over all draws of the validation data set, the mean error on the validation data set equals the expected error.
- Finite sample concentration bounds may be applied to account for finite sample effects.

What about the following for estimating the uncertainty of a model:

- To estimate the uncertainty of a model, we estimate its uncertainty on an i.i.d. calibration data set not used during training.
- In expectation over all draws of the calibration data set, the  $\alpha$ -quantile of uncertainties on the calibration data should be the expected  $\alpha$ -quantile of uncertainty.
- To account for finite sample effects, we apply finite sample concentration bounds.

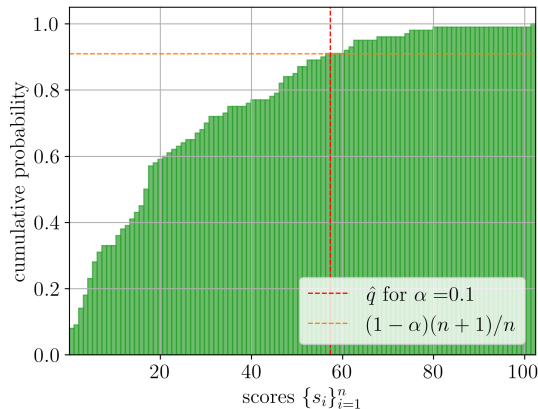
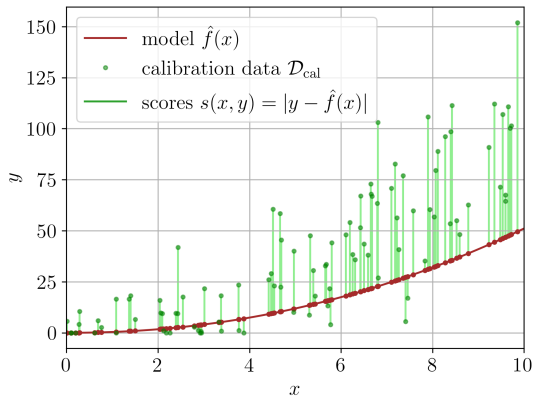
## Basic idea of split conformal prediction II

Let's consider regression with  $\mathcal{Y} = \mathbb{R}$ . The basic split conformal prediction (SCP) approach is rather simple:

1. For each of  $n$  points  $(x_i, y_i) \in \mathcal{D}_{\text{cal}}$  compute a **score function**, say, the absolute error  $s_i = s(x_i, y_i) = |y_i - \hat{f}(x_i)|$ .
2. Sort  $s_1, \dots, s_n$  in increasing order and create an empirical cumulative distribution function (CDF) of errors.
3. Pick an empirical quantile  $\hat{q}$  matching the  $(1 - \alpha)$  coverage.
4. Construct a prediction interval:

$$\mathcal{C}(x_{\text{test}}) = \{y \in \mathcal{Y} \mid s(x_{\text{test}}, y) \leq \hat{q}\} = [\hat{f}(x_{\text{test}}) - \hat{q}, \hat{f}(x_{\text{test}}) + \hat{q}]$$

# Score function and its empirical cumulative distribution



## Split conformal prediction

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### Algorithm 1: Split Conformal Prediction (SCP)

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**Input:** non-conformity score function  $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ ; calibration set

$\mathcal{D}_{cal} = \{(x_i, y_i)\}_{i=1}^n$ ; test point  $x_{test}$ ; target miscoverage level  $\alpha \in (0, 1)$

**Output:** prediction set  $\mathcal{C}(x_{test})$  with coverage  $\geq 1 - \alpha$ , where  $\mathcal{C} : \mathcal{X} \rightarrow 2^{\mathcal{Y}}$

```
1 for  $i = 1$  to  $n$  do  $s_i \leftarrow s(x_i, y_i)$            // Compute scores on calibration set
2 order scores  $s_{(1)} \leq s_{(2)} \leq \dots \leq s_{(n)}$       // Calculate conformal quantile
3 set rank  $k \leftarrow \lceil (1 - \alpha)(n + 1) \rceil$ 
4 if  $k > n$  then set threshold  $\hat{q} \leftarrow \infty$            // Handle edge case
5 else set threshold  $\hat{q} \leftarrow s_{(k)}$ 
6 return  $\mathcal{C}(x_{test}) \leftarrow \{y \in \mathcal{Y} \mid s(x_{test}, y) \leq \hat{q}\}$  // Construct prediction set
```

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## Main theorem: Marginal coverage guarantee

### Theorem (marginal coverage guarantee [VGS99])

Assume that  $\mathcal{C}$  is defined as in SCP (Algorithm 1) for  $\alpha \in (0, 1)$ ,  $\mathcal{D}_{cal} = \{(X_i, Y_i)\}_{i=1}^n$ , and  $\forall_{i \in \{1, \dots, n\}} (X_i, Y_i), (X_{test}, Y_{test}) \stackrel{i.i.d.}{\sim} P$ , where  $P$  is some data distribution. Then

$$1 - \alpha \leq \mathbb{E}_{\mathcal{D}_{cal}} [\mathbb{P}(Y_{test} \in \mathcal{C}(X_{test}) \mid \mathcal{D}_{cal})] \leq 1 - \alpha + \frac{1}{n+1} + \epsilon_{tie} ,$$

where  $\epsilon_{tie}$  captures the likelihood that the nonconformity score of the test data point  $S_{test}$  is the same as another calibration score  $S_i$ ,

$$\epsilon_{tie} := \mathbb{P} \left( \exists_{i \in \{1, \dots, n\}} S_i = S_{test} \right) .$$

## Comments on marginal coverage guarantee

- Bound is pretty tight.
- I.i.d. property can be replaced by a weaker exchangeability property.
- No assumption on loss function.
- Guarantee only holds in expectation over the calibration and test samples.
- We consider  $\hat{q} := \widehat{\mathcal{Q}}_{(1-\alpha)((n+1)/n)}(\{s_i\}_{i=1}^n)$ , which is the  $k$ -th value when sorting  $\{s_i\}_{i=1}^n$  in increasing order with  $k := \lceil (1-\alpha)(n+1) \rceil$ ; this  $k$  can be higher than  $\lceil (1-\alpha)n \rceil$ , that is,  $\widehat{\mathcal{Q}}_{(1-\alpha)}(\{s_i\}_{i=1}^n) \leq \widehat{\mathcal{Q}}_{(1-\alpha)((n+1)/n)}(\{s_i\}_{i=1}^n)$ . This choice allows us to prove the main marginal coverage guarantee later.



## Properties of the residuals

The distribution of the residuals (i.e., the errors of the model) can be categorized as

- Homoscedastic, not dependent on input  $x$ , or
- Heteroscedastic, dependent on input  $x$ .

If  $\mathcal{Y} = \mathbb{R}$  and the prediction set has the form  $[\hat{f}(x) - c_{\text{lo}}, \hat{f}(x) + c_{\text{hi}}]$  then we distinguish (extends canonically to multi-dimensional outputs):

- Symmetric uncertainties, where upper and lower confidence bound are the same.
- Asymmetric uncertainties, where upper and lower confidence bound can differ.

Conformal prediction can be tailored to these settings.

## Examples of score functions for regression

- **Absolute error score (homoscedastic):**  $s(x, y) = |y - \hat{f}(x)|$ , resulting in

$$\mathcal{C}(x) = \left\{ y \in \mathbb{R} \mid |y - \hat{f}(x)| \leq \hat{q} \right\} \iff y \in [\hat{f}(x) - \hat{q}, \hat{f}(x) + \hat{q}] .$$

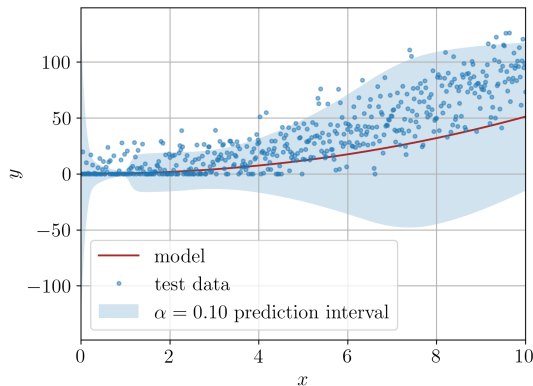
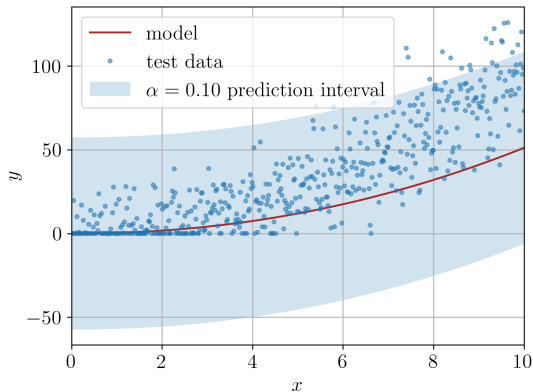
- **Scaled absolute error score (heteroscedastic):** Model also predicts a symmetric uncertainty estimate  $u(x)$ , thus  $s(x, y) = |y - \hat{f}(x)|/u(x)$ , resulting in

$$\mathcal{C}(x) = \left\{ y \in \mathbb{R} \mid \frac{|y - \hat{f}(x)|}{u(x)} \leq \hat{q} \right\} \iff y \in [\hat{f}(x) - \hat{q} \cdot u(x), \hat{f}(x) + \hat{q} \cdot u(x)] .$$

Uncertainty estimates could be

- Predicted variances
- Ensemble variances
- Variances from Bayesian modelling
- ...

## Results for symmetric score functions



**Left:** Absolute error score

**Right:** Scaled absolute error score (using MLP neural network to estimate variance)

## Examples of score functions for asymmetric uncertainties

- **Signed-error Split Conformal Regression [LJL14]:** We use

$$s_{i,\text{lo}} = s_{\text{lo}}(x_i, y_i) = \hat{f}(x_i) - y_i, \quad s_{i,\text{hi}} = s_{\text{hi}}(x_i, y_i) = y_i - \hat{f}(x_i),$$

$$\hat{q}_{\alpha_{\text{lo}}} := \widehat{\mathcal{Q}}_{(1-\alpha_{\text{lo}})((n+1)/n)} \left( \{s_{i,\text{lo}}\}_{i=1}^n \right) \quad \text{and} \quad \hat{q}_{\alpha_{\text{hi}}} := \widehat{\mathcal{Q}}_{(1-\alpha_{\text{lo}})((n+1)/n)} \left( \{s_{i,\text{hi}}\}_{i=1}^n \right),$$

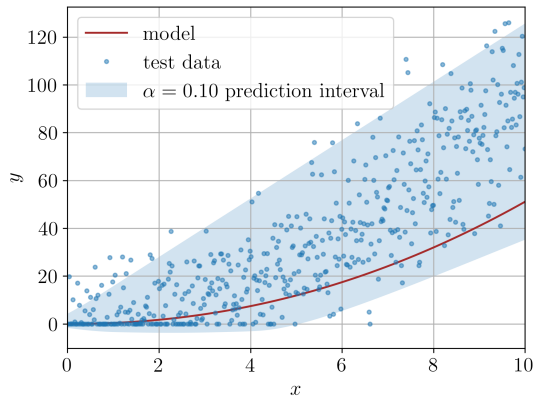
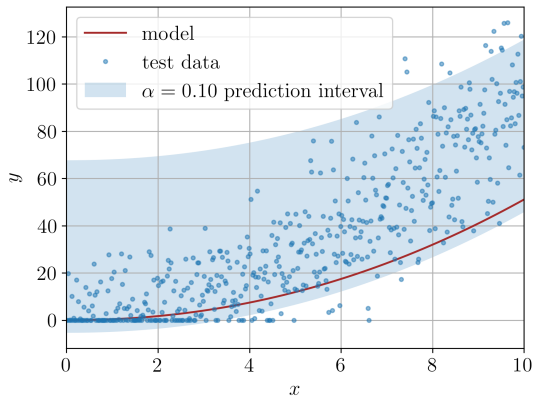
with  $\alpha = \alpha_{\text{lo}} + \alpha_{\text{hi}}$ , giving  $\mathcal{C}(X) = [f(X) - \hat{q}_{\alpha_{\text{lo}}}, f(X_{\text{test}}) + \hat{q}_{\alpha_{\text{hi}}}]$ .

- **Conformalized Quantile Regression (CQR, [RPC19]):** Assuming estimates of lower and upper quantiles  $\hat{q}_{\tau_{\text{lo}}}(x)$  and  $\hat{q}_{\tau_{\text{hi}}}(x)$  at levels  $\tau_{\text{lo}}$  and  $\tau_{\text{hi}}$ , the score function  $s(x, y) = \max \{ \hat{q}_{\tau_{\text{lo}}}(x) - y, y - \hat{q}_{\tau_{\text{hi}}}(x) \}$  corrects both:

$$\begin{aligned} \mathcal{C}(x) &= \{y \in \mathbb{R} \mid \max \{ \hat{q}_{\tau_{\text{lo}}}(x) - y, y - \hat{q}_{\tau_{\text{hi}}}(x) \} \leq \hat{q}\} \\ &\iff y \in [\hat{q}_{\tau_{\text{lo}}}(x) - \hat{q}, \hat{q}_{\tau_{\text{hi}}}(x) + \hat{q}] \end{aligned}$$

The lower and upper quantiles estimates could stem from quantile regression.

## Results of different score functions with asymmetric uncertainties



Left: Signed absolute error score

Right: Conformalized Quantile Regression (using MLPs)

## Properties of common score functions for regression

| Score function                          | heteroscedastic | asymmetric | only requires $\hat{f}(x)$ |
|-----------------------------------------|-----------------|------------|----------------------------|
| Absolute error score                    | no              | no         | yes                        |
| Scaled absolute error score             | yes             | no         | no                         |
| Conformalized Quantile Regression       | yes             | yes        | no                         |
| Signed-error Split Conformal Regression | no              | yes        | yes                        |

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## Order statistics

For a collection of  $n$  random variables  $Z_1, \dots, Z_n$  the  $k$ -th order statistic  $Z_{(k)}$  is the  $k$ -th value when these random variables are arranged in non-decreasing (ascending) order:

$$Z_{(1)} \leq Z_{(2)} \leq \dots \leq Z_{(k)} \leq \dots \leq Z_{(n)} \quad .$$

### Lemma

*For a list of  $n$  i.i.d. random variables  $\mathcal{Z} = (Z_1, \dots, Z_n)$  and any  $k \in \{1, \dots, n\}$  we have*

$$\mathbb{P}\left(Z_i \leq Z_{(k)}\right) \geq \frac{k}{n} \quad \text{and} \quad \mathbb{P}\left(Z_i < Z_{(k)}\right) \leq \frac{k-1}{n} \quad .$$



## Proof of helper lemma

**Proof:** From the definition of order statistics, we have

$$k \leq \sum_{i=1}^n \mathbb{1} \{Z_i \leq Z_{(k)}\}$$

and because expectation preserves inequalities we can rewrite

$$\begin{aligned} k &\leq \mathbb{E} \left[ \sum_{i=1}^n \mathbb{1} \{Z_i \leq Z_{(k)}\} \right] = \sum_{i=1}^n \mathbb{E} [\mathbb{1} \{Z_i \leq Z_{(k)}\}] \\ &= \sum_{i=1}^n \mathbb{P} (Z_i \leq Z_{(k)}) \\ &= n \cdot \mathbb{P} (Z_i \leq Z_{(k)}) \quad \text{for any } i, \end{aligned}$$

because the  $Z_i$  are i.i.d.; dividing both sides by  $n$  proves the first part of the lemma, the second part can be proven analogously.

# Replacement lemma

## Lemma (Replacement lemma)

Let  $S_{(n;k)}$  denote the  $k$ -th order statistic computed over  $\{S_i\}_{i=1}^n$  and  $S_{(n+1;k)}$  the  $k$ -th order statistic computed over  $\{S_i\}_{i=1}^n \cup \{S_{\text{test}}\}$ . Then we have

$$S_{\text{test}} \leq S_{(n;k)} \iff S_{\text{test}} \leq S_{(n+1;k)} .$$

**Proof:**

$S_{\text{test}} \leq S_{(n;k)} \implies S_{\text{test}} \leq S_{(n+1;k)}$ : Proof by contrapositive. Suppose  $S_{\text{test}} > S_{(n+1;k)}$ . Then  $S_{(n+1;k)} = S_{(n;k)}$ , because if we add  $S_{\text{test}}$  and it is strictly greater than  $S_{(n+1;k)}$  its addition did not change the order of the first  $k$  lowest values.

Thus  $S_{\text{test}} > S_{(n+1;k)} \implies S_{\text{test}} > S_{(n;k)}$ .

$S_{\text{test}} \leq S_{(n+1;k)} \implies S_{\text{test}} \leq S_{(n;k)}$ :  $k$ -th smallest entry in the list cannot increase if we add a new value to the list, so  $S_{(n+1;k)} \leq S_{(n;k)}$ .



## Proof of marginal coverage guarantee: Lower bound

$\hat{Q} := \widehat{Q}_{(1-\alpha)((n+1)/n)}(\{S_i\}_{i=1}^n)$  is  $k$ -th largest score  $S_i$  computed over  $n$  samples from  $\mathcal{D}_{\text{cal}}$ , where  $k := \lceil (1 - \alpha)(n + 1) \rceil$ , i.e.,  $\hat{Q} = S_{(k)}$ , and thus

$$\{Y_{\text{test}} \in \mathcal{C}(X_{\text{test}})\} = \{s(X_{\text{test}}, Y_{\text{test}}) \leq \hat{Q}\} = \{S_{\text{test}} \leq S_{(k)}\} \quad ,$$

where  $S_{\text{test}} := s(X_{\text{test}}, Y_{\text{test}})$ .

$\mathcal{D}_{\text{cal}}$  and  $(X_{\text{test}}, Y_{\text{test}})$  being i.i.d. implies that the scores  $S_i$  and  $S_{\text{test}}$  are i.i.d.

Combining both lemmata and choosing  $k = \lceil (1 - \alpha)(n + 1) \rceil$  gives

$$\mathbb{P}(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}})) \stackrel{\text{replacement lemma}}{=} \mathbb{P}(S_{\text{test}} \leq S_{(n+1;k)}) \geq \frac{k}{n+1} = \frac{\lceil (1 - \alpha)(n + 1) \rceil}{n+1} \geq 1 - \alpha \quad ,$$

using the simplified notation  $\mathbb{P}(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}})) := \mathbb{E}_{\mathcal{D}_{\text{cal}}}[\mathbb{P}(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}}) \mid \mathcal{D}_{\text{cal}})]$ .

## Proof of marginal coverage guarantee: Upper bound

From

$$S_{\text{test}} \leq S_{(n+1;k)} \iff \text{either } S_{\text{test}} < S_{(n+1;k+1)} \text{ or } S_{\text{test}} = S_{(n+1;k)} = S_{(n+1;k+1)}$$

we have

$$\begin{aligned} \mathbb{P}(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}})) &= \mathbb{P}(S_{\text{test}} \leq S_{(n+1;k)}) \\ &= \mathbb{P}(S_{\text{test}} < S_{(n+1;k+1)}) + \mathbb{P}(S_{\text{test}} = S_{(n+1;k)} = S_{(n+1;k+1)}) \\ &\leq \mathbb{P}(S_{\text{test}} < S_{(n+1;k+1)}) + \epsilon_{\text{tie}} \\ &\leq \frac{(k+1)-1}{n+1} + \epsilon_{\text{tie}} = \frac{\lceil (1-\alpha)(n+1) \rceil}{n+1} + \epsilon_{\text{tie}} \\ &\leq 1 - \alpha + \frac{1}{n+1} + \epsilon_{\text{tie}} , \end{aligned}$$

where the second equality holds since the two events are mutually exclusive.



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## Calibration-conditional coverage theorem

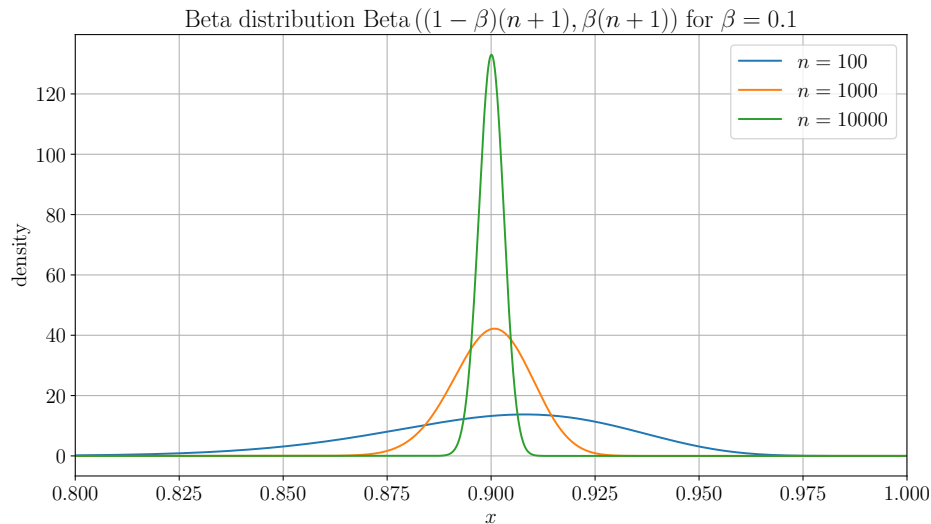
### Theorem (Calibration-conditional coverage)

*Suppose the data points  $\{(X_i, Y_i)_i\}_{i=1}^n = \mathcal{D}_{cal}$  and  $(X_{test}, Y_{test})$  are i.i.d., and let  $\mathcal{C}$  be constructed via SCP (Algorithm 1) using any pre-trained non-conformity score function  $s$  at a miscoverage level  $\beta$ . Then for any  $\alpha \in (\beta, 1)$  the calibration-conditional coverage stochastically dominates the beta distribution*

$$\mathbb{P}(\mathbb{P}(Y_{test} \in \mathcal{C}(X_{test}) \mid \mathcal{D}_{cal}) \leq 1 - \alpha) \leq F_{\text{Beta}((1-\beta)(n+1), \beta(n+1))}(1 - \alpha) ,$$

*where  $F_{\text{Beta}(a,b)}$  denotes the CDF of the Beta  $(a, b)$  distribution.*

# Beta distribution



## Calibration-conditional coverage theorem

**Proof:** Let  $F_S$  be the CDF of the distribution of scores  $s(X, Y)$ . Define  $S_i = s(X_i, Y_i)$ , which are i.i.d. draws from the distribution with CDF  $F_S$ . Let  $S_{(1)} \leq \dots \leq S_{(n)}$  be the order statistics of  $S_1, \dots, S_n$ . For  $k = \lceil (1 - \beta)(n + 1) \rceil$ , we established

$$\mathbb{P}(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}}) \mid \mathcal{D}_{\text{cal}}) = \mathbb{P}(S_{\text{test}} \leq S_{(k)} \mid \mathcal{D}_{\text{cal}}) = F_S(S_{(k)}) \quad ,$$

where the last step holds since  $S_{\text{test}}$  is independent of  $\mathcal{D}_{\text{cal}}$  and has CDF  $F_S$ . Therefore

$$\mathbb{P}(\mathbb{P}(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}}) \mid \mathcal{D}_{\text{cal}}) \leq 1 - \alpha) = \mathbb{P}(F_S(S_{(k)}) \leq 1 - \alpha) \quad .$$

Let  $Z_i := F_S(S_i)$ . Since  $F_S$  is non-decreasing, the order statistics  $Z_{(1)} \leq \dots \leq Z_{(n)}$  of  $Z_1, \dots, Z_n$  satisfy  $Z_{(k)} = F_S(S_{(k)})$ .



## Calibration-conditional coverage theorem

By the probability integral transform, if  $F_S$  is continuous, each  $Z_i$  is an i.i.d. sample from the uniform distribution  $\mathcal{U}_{[0,1]}$ . Otherwise, if  $F_S$  has discontinuities, the resulting  $Z_i$  stochastically dominate  $U_1, \dots, U_n \sim \mathcal{U}_{[0,1]}$ , meaning

$$\forall_{i \in [n], x \in [0,1]} F_{Z_i}(x) \leq F_{U_i}(x),$$

we get

$$\mathbb{P}(F_S(S_{(k)}) \leq 1 - \alpha) = \mathbb{P}(Z_{(k)} \leq 1 - \alpha) \leq \mathbb{P}(U_{(k)} \leq 1 - \alpha) .$$

By the definition of the beta distribution, the  $k$ -th order statistic of a sample of size  $n$  from  $\mathcal{U}_{[0,1]}$  is a beta random variable

$$U_{(k)} \sim \text{Beta}(k, n + 1 - k) ,$$

and since  $k \geq (1 - \beta)(n + 1)$ ,  $F_S(S_{(k)})$  stochastically dominates  $\text{Beta}((1 - \beta)(n + 1), \beta(n + 1))$ .



## Calibration-conditional coverage upper bound

### Lemma (Calibration-conditional coverage upper bound)

*Under the conditions of the **calibration-conditional coverage** theorem, we have that*

$$\mathbb{P}(\mathbb{P}(Y_{test} \in \mathcal{C}(X_{test}) \mid \mathcal{D}_{cal}) \leq 1 - \alpha) \leq e^{-2n\Delta^2},$$

*where  $\beta := \alpha - \Delta$ .*

## Proof of calibration-conditional coverage upper bound

**Proof:** In the proof of the **calibration-conditional coverage** theorem, we established that

$$\mathbb{P}(\mathbb{P}(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}}) \mid \mathcal{D}_{\text{cal}}) \leq 1 - \alpha) \leq \mathbb{P}(U_{(k)} \leq 1 - \alpha) ,$$

where  $U_{(k)}$  is the  $k$ -th order statistic of  $n$  uniformly distributed random variables  $U_1, \dots, U_n \sim \mathcal{U}_{[0,1]}$  with  $k = \lceil (1 - \beta)(n + 1) \rceil = \lceil (1 - \alpha + \Delta)(n + 1) \rceil$ . This is equivalent to saying that at least  $k$  realizations of  $U_i$  are  $\leq 1 - \alpha$ . Thus, by using  $V_i = \mathbb{1}\{U_i \leq 1 - \alpha\}$  the probability can be rewritten as

$$\mathbb{P}(U_{(k)} \leq 1 - \alpha) = \mathbb{P}\left(\sum_{i=1}^n V_i \geq k\right) \leq \mathbb{P}\left(\sum_{i=1}^n V_i \geq (1 - \alpha + \Delta)(n + 1)\right) .$$

## Proof of calibration-conditional coverage upper bound

Since  $V_1, \dots, V_n$  are i.i.d. with expected values

$$\mathbb{E}[V_i] = \mathbb{E}[\mathbb{1}\{U_i \leq 1 - \alpha\}] = \mathbb{P}(U_i \leq 1 - \alpha) = 1 - \alpha$$

we can apply Hoeffding's inequality:

$$\begin{aligned} \mathbb{P}\left(\sum_{i=1}^n V_i \geq (1 - \alpha + \Delta)(n + 1)\right) &\leq \mathbb{P}\left(\sum_{i=1}^n V_i \geq (1 - \alpha + \Delta)n\right) \\ &= \mathbb{P}\left(\sum_{i=1}^n V_i - n\mathbb{E}[V] \geq (1 - \alpha + \Delta)n - n(1 - \alpha)\right) \\ &= \mathbb{P}\left(\sum_{i=1}^n V_i - n\mathbb{E}[V] \geq n\Delta\right) \\ &\leq e^{-2n\Delta^2} \end{aligned}$$

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## Bin-wise conditional coverage (poor man's conditional coverage)

### Theorem (Bin-wise conditional coverage)

Assuming the setup of the *marginal coverage guarantee* theorem, partition  $\mathcal{X}$  into  $B$  bins  $\mathcal{X}_1, \dots, \mathcal{X}_B$ . Further, let  $\mathcal{I}_b = \{i \in \{1, \dots, n\} \mid X_i \in \mathcal{X}_b\}$  be the index set of features  $X_i$  belonging to bin  $\mathcal{X}_b$ ,  $n_b = |\mathcal{I}_b|$  and  $\hat{q}_b := \widehat{Q}_{(1-\alpha)((n_b+1)n_b)}(\{S_i\}_{i \in \mathcal{I}_b})$  be the empirical quantile computed according to Algorithm (1) for each bin  $b \in \{1, \dots, B\}$  separately by partitioning  $\mathcal{D}_{cal}$  into  $\{(X_i, Y_i)\}_{i \in \mathcal{I}_b}$ . By defining

$$\mathcal{C}(X_{test}) = \{y : X_{test} \in \mathcal{X}_m \wedge S_{test} \leq \hat{q}_b\}$$

we get the bin-wise conditional coverage guarantee

$$\mathbb{P}(Y_{test} \in \mathcal{C}(Y_{test}) \mid X_{test} \in \mathcal{X}_b) \geq 1 - \alpha$$

for all  $b \in \{1, \dots, B\}$  with  $\mathbb{P}(X_{test} \in \mathcal{X}_b) > 0$ .

## Split Localized Conformal Prediction (SLCP)

- To get test-conditional coverage, we would ideally like to know the conditional CDF

$$q_{1-\alpha}^*(x) := \mathcal{Q}_{1-\alpha} \left( F_{S|X}(S|x) \right)$$

with

$$\mathbb{P}(S \leq q_{1-\alpha}^*(x) \mid X = x) = 1 - \alpha \ .$$

- **Idea [HTGL23]:** Approximate  $q_{1-\alpha}^*(x)$  by  $\tilde{q}_{1-\alpha}(x) \approx q_{1-\alpha}^*(x)$ , consider

$$\mathbb{P}(S - \tilde{q}_{1-\alpha}(x) \leq q_{1-\alpha}^*(x) - \tilde{q}_{1-\alpha}(x) \mid X = x) = 1 - \alpha \ ,$$

and apply conformal prediction on the residual of ideal and approximated quantiles

$$\mathbb{P}(S' \leq \hat{q}) \geq 1 - \alpha \ ,$$

with the new score  $S' = S - \tilde{q}_{1-\alpha}(x)$ .

## Prediction interval using absolute error score

Let us assume that the original score function is the absolute error:

$$\mathcal{C}(x) = \left\{ y \in \mathbb{R} \mid \overbrace{|y - \hat{f}(x)| - \tilde{q}_{1-\alpha}(x)}^{s(x,y)} \leq \hat{q} \right\}$$

$$\iff y \in \left\{ \hat{f}(x) - \underbrace{\hat{q}}_{\substack{\text{CP} \\ \text{correction}}} - \underbrace{\tilde{q}_{1-\alpha}(x)}_{\substack{\text{estimated} \\ \text{quantile}}}, \hat{f}(x) + \hat{q} + \tilde{q}_{1-\alpha}(x) \right\}$$



## Example using Nadaraya–Watson estimator and signed absolute error

- **Nadaraya–Watson estimator** for conditional CDF: Given  $\{(x_i, y_i)\}_{i=1}^m$ , kernel  $K : \mathbb{R} \rightarrow \mathbb{R}$  with zero mean, the Nadaraya–Watson estimator of the CDF  $F_{Y|X}$  is

$$\hat{F}_{Y|X}(y|x) = \sum_{i=1}^m w(x_i|x) \mathbb{1}\{y \leq y_i\}$$

with

$$w(x_i|x) = \frac{K(\|x_i - x\|)}{\sum_{j=1}^m K(\|x_j - x\|)} .$$

- We estimate the localized conditional CDF **on the training data** ( $\mathcal{D}_{\text{cal}}$  and test data remain exchangeable):

$$\tilde{q}_{1-\alpha}(x) = \tilde{\mathcal{Q}}_{1-\alpha}(\hat{F}_{S|X}(S|x)) = \inf \left\{ s : 1 - \alpha \leq \sum_{(x', y') \in \mathcal{D}_{\text{train}}} w(x'|x) \mathbb{1}\{s \leq s(x', y')\} \right\} .$$

- We use Signed-error Split Conformal Regression.

# SLCP with kernel density estimation

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## Algorithm 2: Split Localized Conformal Prediction (SLCP)

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**Input:**  $\mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{train}} \in 2^{\mathcal{X} \times \mathcal{Y}}$ ; model  $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$ ; test input  $x_{\text{test}} \in \mathcal{X}$ ;  $\alpha_{\text{lo}}, \alpha_{\text{hi}} \in (0, 1)$ ;  
kernel  $K : \mathbb{R} \rightarrow \mathbb{R}$

**Output:** Prediction set  $\mathcal{C}(x_{\text{test}}) \in 2^{\mathcal{Y}}$  with coverage  $\geq 1 - (\alpha_{\text{lo}} + \alpha_{\text{hi}})$  and  
 $\forall_{(x,y) \in \mathcal{X} \times \mathcal{Y}} \lim_{|\mathcal{D}_{\text{cal}}| \rightarrow \infty} \mathbb{P}(y \in \mathcal{C}(x) | X_{\text{test}} = x) \geq 1 - (\alpha_{\text{lo}} + \alpha_{\text{hi}}).$

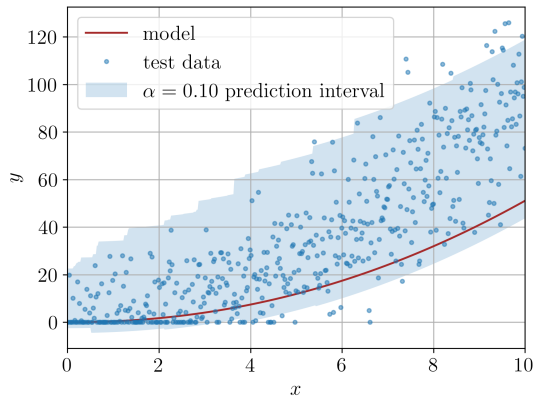
```

1  $s_{\text{lo}}(x, y) := -s_{\text{hi}}(x, y) := \hat{f}(x) - y$ 
2 for  $l \in \{\text{lo}, \text{hi}\}$  do
3    $w(x_i | x) := \frac{K(\|x_i - x\|)}{\sum_{(x_j, y_j) \in \mathcal{D}_{\text{train}}} K(\|x_j - x\|)}$ 
4    $\tilde{q}_{1-\alpha_l}(x) := \inf \left\{ s : 1 - \alpha_l \leq \sum_{(x_i, y_i) \in \mathcal{D}_{\text{train}}} w(x_i | x) \mathbb{1}[s \leq s(x_i, y_i)] \right\}$ 
5    $\hat{q}_l \leftarrow$  run SCP on  $\mathcal{D}_{\text{cal}}$  with miscoverage  $\alpha_l$  and score  $s(x, y) = s_l(x, y) - \tilde{q}_{1-\alpha_l}(x)$ 
6 return
```

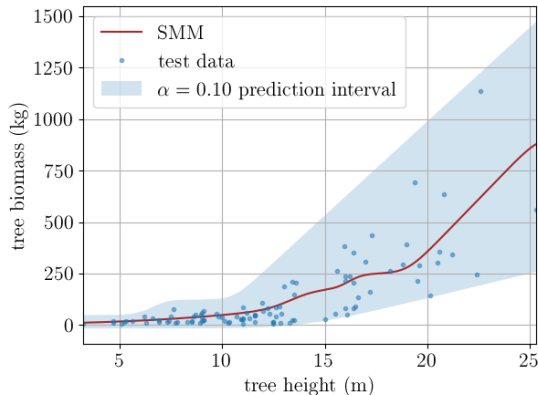
$\mathcal{C}(x_{\text{test}}) := \left\{ y \in \mathcal{Y} \mid y \in \left[ \hat{f}(x_{\text{test}}) - \tilde{q}_{1-\alpha_{\text{lo}}}(x_{\text{test}}) - \hat{q}_{\text{lo}}, \hat{f}(x_{\text{test}}) + \tilde{q}_{1-\alpha_{\text{hi}}}(x_{\text{test}}) + \hat{q}_{\text{hi}} \right] \right\}$

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# SLCP results



## Real-world example



- Data:** Allometric data from trees in southern China (Qiu et al., in preparation)
- Model:** Smooth min-max (SMM) neural network (Igél, *ICML*, 2024)
- CP method:** CQR with MLPs

## Summary

### Conformal prediction

- gives prediction sets with rigorous coverage guarantees,
- requires no assumption on model, data generating distribution, and algorithm except calibration data and test data being i.i.d.,
- can deal with heteroscedastic, asymmetric noise/uncertainty,
- can turn your favourite uncertainty estimates into proper prediction sets.

Conditional coverage remains difficult.

Conformal prediction goes back to Vovk and Gammerman (e.g., [VGS99]), there is an excellent review by Angelopoulos and Bates [AB23], and a new textbook draft by Angelopoulos, Barber, and Bates [ABB25] – we based this tutorial on the latter.



# Thanks



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## Hoeffding's inequality

### Theorem (Hoeffding's inequality)

*Let  $X_1, \dots, X_n$  be independent real-valued random variables, such that for each  $i \in \{1, \dots, n\}$  there exist  $a_i \leq b_i$ , such that  $X_i \in [a_i, b_i]$ . Then for every  $\epsilon > 0$ :*

$$\mathbb{P} \left( \sum_{i=1}^n X_i - \mathbb{E} \left[ \sum_{i=1}^n X_i \right] \geq \epsilon \right) \leq e^{-2\epsilon^2 / \sum_{i=1}^n (b_i - a_i)}$$

*and*

$$\mathbb{P} \left( \mathbb{E} \left[ \sum_{i=1}^n X_i \right] - \sum_{i=1}^n X_i \geq \epsilon \right) \leq e^{-2\epsilon^2 / \sum_{i=1}^n (b_i - a_i)} .$$

# Probability integral transform

## Theorem (Probability integral transform)

*Suppose that a random variable  $X$  has a continuous distribution for which the cumulative distribution function (CDF) is  $F_X$ . Then the random variable  $Y$  defined as*

$$Y := F_X(X)$$

*has a standard uniform distribution  $\mathcal{U}_{[0,1]}$ . If  $F_X$  is not continuous then  $Y$  is a superuniform random variable, meaning that*

$$\forall_{u \in [0,1]} \mathbb{P}(Y \leq u) \leq u \ ,$$

*or equivalently,  $Y$  dominates the uniform distribution:*

$$\forall_{u \in [0,1]} F_Y(u) \leq F_U(u) \ , \quad \text{where} \quad U \sim \mathcal{U}_{[0,1]} \ .$$